

1

NUMBER AND ALGEBRA

ALGEBRA

Algebra uses symbols and formulas to describe number patterns and relationships in our natural and physical world. Algebra has helped people to unlock some of the secrets of the universe, such as showing that mass and energy are related to one another by the equation $E = mc^2$. Algebra is used to better understand climate change, population growth and economic growth. Algebraic modelling is used to predict the behaviour of bushfires, which may help save lives and reduce property damage.



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Chapter outline

	Working mathematically				
1.01 From words to algebraic expressions	U	F		R	C
1.02 Substitution	U	F			
1.03 Adding and subtracting terms	U	F		R	
1.04 Multiplying and dividing terms	U	F		R	
1.05 Adding and subtracting algebraic fractions	U	F			
1.06 Multiplying and dividing algebraic fractions	U	F			
1.07 Expanding expressions	U	F			
1.08 Factorising expressions	U	F			
1.09 Expanding binomial products	U	F		R	C

Wordbank

Quiz
Wordbank 1

binomial An algebraic expression that consists of 2 terms, for example, $4a + 9$, $3 - y$, $x^2 - 4x$

consecutive numbers Any series of integers that follow each other in order, for example, 8, 9 and 10

evaluate To find the value of an algebraic expression or formula

expand To rewrite an expression such as $5(2k - 6)$ without grouping symbols; for example, $5(2k - 6)$ expands to $10k - 30$

factorise To rewrite an expression with grouping symbols, by taking out the highest common factor; factorising is the opposite of expanding; for example, $9r^2 + 36r$ factorised is $9r(r + 4)$

highest common factor (HCF) The largest term that is a factor of 2 or more terms, for example, the HCF of $9r^2 + 36r$ is $9r$

simplify To make shorter or less complex, for example, $4a + 3a$ simplifies to $7a$

substitute To replace a variable with a number in an algebraic expression or formula

variable (or **pronumeral**) A symbol, such as a, b or c used to represent numbers

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (10):

1.03 Adding and subtracting terms
1.04 Multiplying terms • Dividing terms
1.05 Adding and subtracting algebraic fractions
1.07 Expanding expressions 1 • Expanding expressions 2
1.08 HCF of algebraic terms • Factorising expressions
1.09 Expanding binomial products 1 • Expanding binomial products 2

Twig video (1):

1.02 Heptathlon

PhET interactives (2):

1.03 Expression exchange
1.07 Area model Algebra

Quizzes (6):

- Wordbank 1
- SkillCheck 1
- Mental skills 1A
- Mental skills 1B
- Language of maths 1
- Test yourself 1

Skillsheets (5):

1.01 Algebraic expressions
1.02 Spreadsheets
1.03, 1.07 Algebra using diagrams
1.08 HCF by factor trees • Factorising using diagrams

Worksheets (5):

1.04 Perimeter and area
1.07 Algebra review
1.09 Area diagrams • Binomial products
Mind map Algebra

Puzzles (8):

1.01 Generalised arithmetic
1.02 Substitution puzzle
1.05 Algebraic fractions
1.06 Algebraic fractions puzzle • Upside-down fractions
1.08 Factorising puzzle
1.09 Expanding brackets
Language of maths Algebra crossword

Technology (1):

1.09 Expanding binomials

Spreadsheet (1):

1.09 Expanding binomials

Presentation (1):

1.05 Adding and subtracting fractions

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- convert between worded and algebraic expressions
- substitute values to evaluate algebraic expressions
- add, subtract, multiply and divide algebraic expressions
- add, subtract, multiply and divide simple algebraic fractions
- expand algebraic expressions
- factorise algebraic expressions
- expand binomial products

SkillCheck ANSWERS ON P. 562

1 Write an algebraic expression for each statement.

a The sum of m and n **b** The product of n and m

c Increase n by 5 **d** Decrease 5 by n

2 If $p = 4$, find the value of each expression.

a $p + 6$ **b** $3p$ **c** p^2 **d** $10p^2$

3 State whether each pair of terms are like or unlike terms.

a $2y$ and $5y$ **b** $8ab$ and $7ba$ **c** $6p$ and $-p$

d $3x^2$ and $3y^2$ **e** $5m^2$ and $3m$ **f** $4w^3y$ and $12w^2y^2$

4 Simplify each expression.

a $5n + 6n$ **b** $9h - 3h$ **c** $10pq - qp$ **d** $x^2 + 4x^2$

e $4 \times w$ **f** $8 \times 5x$ **g** $2g \times 3h$ **h** $3y \times 6y$

i $10x \div 2$ **j** $6b \div b$ **k** $\frac{16xy}{8x}$ **l** $\frac{15n^2}{25n}$

5 Evaluate each expression.

a $\frac{2}{5} + \frac{1}{3}$ **b** $\frac{3}{4} - \frac{2}{5}$ **c** $\frac{5}{8} \times \frac{1}{2}$ **d** $\frac{4}{5} \div \frac{2}{7}$

e $\frac{6}{11} \times \frac{5}{12}$ **f** $\frac{7}{8} + \frac{1}{3}$ **g** $\frac{4}{9} \div \frac{3}{10}$ **h** $\frac{5}{6} - \frac{3}{4}$

Quiz
SkillCheck 1

From words to algebraic expressions 1.01

In algebra, letters of the alphabet are used to represent numbers. Such letters are called **variables**. For example, a general way of stating 'any number plus 5' is the **algebraic expression** '$n + 5$' where the variable n stands for 'any number'.

- n is called a variable because its value can change (vary)
- n is also called a **pronumeral** because it stands for a number

Variables are useful for writing general rules or formulas.

For example, the average of the 2 numbers 7 and 11 is $\frac{7+11}{2} = 9$. In general, we can say that the average of *any* 2 numbers, x and y, is the algebraic expression $\frac{x+y}{2}$.

Puzzle
Generalised arithmetic

Skillsheet
Algebraic expressions

Example 1

Convert each statement to an algebraic expression.

a The difference between 2 numbers m and n.

b The **product** of 4 and a number h.

Solution

a $m - n$ Difference is the answer to a subtraction

b $4 \times h = 4h$ Product is the answer to a multiplication

Example 2

Write an algebraic expression for the result of each statement.

a Think of a number (n), multiply it by 3 and then subtract 5.

b Think of a number (n), subtract 5 and then multiply by 3.

c Think of a number (n), divide by 2 and then add 7.

d Think of a number (n), add 7 and then divide by 2.

Solution

a $n \times 3 - 5 = 3n - 5$

b $(n - 5) \times 3 = 3(n - 5)$

c $n \div 2 + 7 = \frac{n}{2} + 7$

d $(n + 7) \div 2 = \frac{n+7}{2}$

Example 3

a Find the number of hours in:

i 180 minutes **ii** t minutes.

b A mobile phone plan costs \$24 per month plus \$10/GB for excess data use over 8 GB. Find the total cost of the plan for:

i 12 months and 10 GB of data used per month

ii w months and 15 GB of data used per month

Solution

a i Number of hours $= 180 \div 60$
$= 3$

ii Number of hours $= t \div 60$
$= \frac{t}{60}$

b i Total cost $= 12 \times (\$24 + 2 \times \$10)$ 10 GB is 2 GB over 8 GB
$= \$528$

ii Total cost $= w \times (\$24 + 7 \times \$10)$ 15 GB is 7 GB over 8 GB
$= \$94w$

EXERCISE 1.01 ANSWERS ON P. 562

From words to algebraic expressions U F R C

1 Match each worded statement from column A with the correct algebraic expression from column B if x stands for 'a number'.

EXAMPLE 1

Column A		Column B	
a	The sum of a number and 8.	**A**	$2x$
b	Divide a number by 6.	**B**	$x - 11$
c	11 less than a number.	**C**	$5x$
d	The product of a number and 7.	**D**	$x - 6$
e	Increase 20 by a number.	**E**	$x + 13$
f	Multiply a number by 3.	**F**	$\frac{20}{x}$
g	The difference between a number and 6.	**G**	$20 - x$
h	Divide 20 by a number.	**H**	$20 + x$
i	Decrease a number by 2.	**I**	$3x$
j	20 decreased by a number.	**J**	$30 - x$
k	13 more than a number.	**K**	$x - 30$
l	A number multiplied by itself.	**L**	$x + 8$
m	Subtract seven from a number.	**M**	$\frac{x}{6}$
n	Double a number.	**N**	$x - 2$
o	Remove a number from 30.	**O**	$x - 7$
p	30 less than a number.	**P**	$7x$
q	5 lots of a number.	**Q**	x^2

2 Write an algebraic expression for each statement. C

a 5 more than x.
b 3 less than m.
c The product of 4 and k.
d 20 divided by w.
e Increase 7 by d.
f The product of h and k.
g The difference between $9g$ and $4y$.
h $5a$ divided by 4.
i Decrease $3d$ by $2w$.
j The square of k.
k How many times r goes into t.
l Subtract $7p$ from $2q$.
m 2 less than double a.
n 5 more than half d.
o One-third of the product of g and h.
p Three times n less seven.
q 4 less than twice n
r 10 more than negative b.

3 Write an algebraic expression for the result of each statement. C

EXAMPLE 2

a Think of a number (t), multiply it by 3 and then add 5.
b Think of a number (t), add 5 and then multiply by 3.
c Think of a number (w), multiply it by 7 and then subtract 4.
d Think of a number (w), subtract 4 and then multiply by 7.

Foundation | Standard | Complex

e Think of a number (k), add 8 and then divide by 5.

f Think of a number (k), divide by 5 and then add 8.

g Think of a number (d), divide by 6 and then subtract 2.

h Think of a number (d), subtract 2 and then divide by 6.

4 Write each algebraic expression in words. C

a $b + 5$ **b** $5b$ **c** $7 - b$ **d** $\frac{b}{3}$

e $2b + 3$ **f** $4 - 3b$ **g** $5 + \frac{b}{4}$ **h** $2b^2$

EXAMPLE 3

5 If Rimsha is 16 years old, use an algebraic expression to write her age: R C

a in y years time **b** k years ago.

6 The suggested cooking time for pork in a moderately hot oven is t minutes per kilogram of pork plus 15 minutes extra (for the crackling).

Which expression gives the cooking time for a 3 kg leg of pork? Select the correct answer **A**, **B**, **C** or **D**.

A $3(t + 15)$ **B** $\frac{t}{3} + 15$

C $t + 18$ **D** $3t + 15$



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7 The cost of a taxi trip is \$3.80 plus \$2.30/km. Find the total cost of a taxi trip for:

a 12 km **b** D km.

8 The cost of an international phone call to Italy is a connection fee of \$0.45 plus \$0.80 per minute. Which expression gives the cost of such a call lasting k minutes? Select **A**, **B**, **C** or **D**.

A $\$(0.45 + 80k)$ **B** $\$(0.45 + 0.8k)$

C $\$(45 + 0.8k)$ **D** $\$(0.8 + 45k)$

9 Write an algebraic expression for each statement. C

a The cost of p books at \$$Q$ each.

b The selling price after a GST of \$$t$ is added to the marked price of \$40.

c The number of metres in h km.

d The average of m, p, v and w.

e The number of dollars in y cents.

f The discount when a \$950 large screen TV sells for a price of \$$x$.

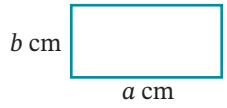
g The change in dollars from \$100 after buying w tickets at \$12 each.

h The amount of each share when \$450 is divided equally among n people.

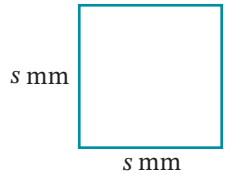
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i The perimeter of this rectangle.



j The area of this square.



10 **a** Two angles of a triangle are 130° and 25°. Find the size of the third angle.

b Two angles of a triangle have size $a°$ and $b°$. Find an algebraic expression for the size of the third angle.

11 **a** The unequal angle in an isosceles triangle is 70°. Find the size of one of the equal angles.

b The unequal angle in an isosceles triangle is $x°$. Find an algebraic expression for the size of one of the equal angles. R C

12 If d is an even number, what is the next: R C

a even number? **b** odd number?

13 **Consecutive numbers** are integers that follow one another, such as 3, 4, 5 and 6. What is the next consecutive number after:

a 11? **b** −9? **c** x? **d** $n + 4$?

14 Write 4 consecutive numbers beginning with: R C

a 8 **b** −3 **c** $y + 4$ **d** $n - 4$

15 Write 4 consecutive *odd* numbers beginning with: R C

a 5 **b** 29 **c** t **d** $k - 1$

Substitution 1.02

When the variables in an algebraic expression are replaced with numbers, this is called **substitution**. When the value of the algebraic expression is then calculated, it is called **evaluation**.

E-*valu*-ate means 'find the value of'.



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Puzzle
Substitution puzzle

Video
Heptathlon

Foundation Standard Complex

Example 4

If $f = 5$, $g = -4$ and $r = 2$, evaluate each of the following expressions.

a $2f + g$ **b** $8 + rf$ **c** $f^2 - r^2$

d $3g^2 - 5$ **e** $r(f - g)$ **f** $\frac{5r + f}{3}$

Solution

a $2f + g = 2 \times 5 + (-4)$
$= 6$

b $8 + rf = 8 + 2 \times 5$
$= 18$

c $f^2 - r^2 = 5^2 - 2^2$
$= 21$

d $3g^2 - 5 = 3 \times (-4)^2 - 5$
$= 43$

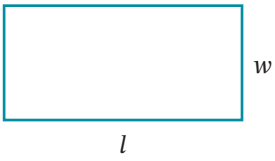
e $r(f - g) = 2 \times [5 - (-4)]$
$= 18$

f $\frac{5r + f}{3} = \frac{5 \times 2 + 5}{3}$
$= 5$

Substituting into formulas

A **formula** is an algebraic rule that shows the relationship between variables. For example, if a rectangle has length l and width w, then:

- the formula $P = 2l + 2w$ gives the perimeter of the rectangle
- the formula $A = lw$ gives the area of the rectangle.



Note: $3g^2 - 5$ is called an 'expression' but $P = 2l + 2w$ is called an 'equation' or 'formula' because it has an equals sign.

Example 5

A rectangle has length 3.4 metres and width 2.5 metres.

Use the formula $P = 2l + 2w$ to calculate its perimeter.

Solution

$l = 3.4$, $w = 2.5$

$P = 2l + 2w$
$= 2 \times 3.4 + 2 \times 2.5$
$= 11.8$

The perimeter is 11.8 metres.

EXERCISE 1.02 ANSWERS ON P. 562

Substitution U F

EXAMPLE 4

1 Find the value of $7a + 3$ if:

a $a = 7$ **b** $a = 15$ **c** $a = -3$ **d** $a = -20$

2 Find the value of w^2 if:

a $w = 6$ **b** $w = 15$ **c** $w = -8$ **d** $w = -15$

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3 Evaluate $8 - 3x$ if:

a $x = 12$ **b** $x = 8$ **c** $x = 2.5$ **d** $x = -10$

4 Evaluate $\frac{h}{2} + 7$ if:

a $h = 2$ **b** $h = 7$ **c** $h = -14$ **d** $h = -30$

5 Evaluate each expression.

a $2a + b$, when $a = 3$ and $b = 1.5$

b $3k - \frac{d}{3}$, when $k = 10$ and $d = 12$

c $\frac{f}{5} + f$, when $f = 20$

d $3x + 2y$, when $x = -4$ and $y = 7$

e $\frac{m+n}{2}$, when $m = 7$ and $n = 11$

f $n^2 - 3n$, when $n = -6$

6 Evaluate each expression if $m = 2$, $n = -3$ and $p = 5$.

a $m + n + p$ **b** $m - n$ **c** np

d $m^2 - n^2$ **e** $np - m$ **f** $mn + p$

g $5n + 3p$ **h** $\sqrt{m+n+p}$ **i** $\frac{p+n}{m}$

j $5m^2$ **k** $m^2 + p - 4n$ **l** $\frac{p}{n-m}$

7 If the perimeter of a rectangle with length l and width w has the formula $P = 2l + 2w$, find the perimeter of a rectangle with length 4.7 cm and width 2.5 cm.

8 If the area of a rectangle has the formula $A = lw$, find the area of a rectangle with length 12.5 m and width 8 m.

9 If the volume of a rectangular pyramid with base area A, and height h is $V = \frac{1}{3}Ah$, find the volume of the rectangular prism with base area 28 cm^2 and height 30 cm.

10 The area of a triangle with base length b and perpendicular height h is $A = \frac{1}{2}bh$.
Find the area of a triangle with base length 15 cm and perpendicular height 12 cm.

11 Given that $A = \frac{h(a+b)}{2}$, evaluate A when $h = 4.5$, $a = 6$ and $b = 8$.
Select the correct answer **A**, **B**, **C** or **D**.

A 17.5 **B** 31.5 **C** 63 **D** 108

12 If $M = \frac{x+y+z}{3}$, find M when $x = 3.7$, $y = 8.1$ and $z = 0.5$.

13 Given $m = \frac{5k}{8}$, find m when $k = 50$.

14 Given that $P = -2d^2 - 5d + 8$, find P when $d = -4$. Select **A**, **B**, **C** or **D**.

A 60 **B** 4 **C** −4 **D** 50

1.02

EXAMPLE 5

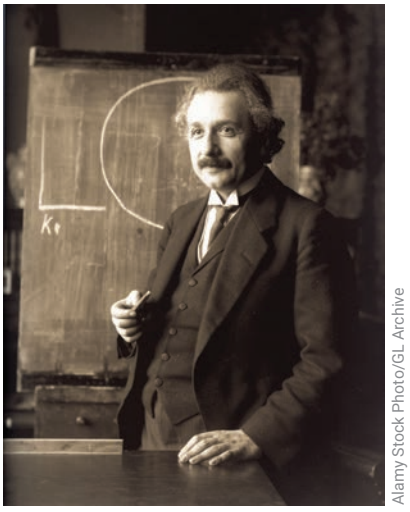
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Did you know?

Einstein's formula

In 1905, when the German scientist Albert Einstein was only 26, he proposed a new theory of physics for small particles of matter (such as atoms) moving at very high speeds. He proposed that matter (mass) could be converted into large amounts of energy, describing this in his Special Relativity formula $E = mc^2$, where E stands for energy, m stands for mass and c stands for the speed of light (3 billion metres per second). Einstein's theory revolutionised conventional laws of physics and led to the development of nuclear energy.

Find out why Einstein renounced his German citizenship and became an American in 1940.



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Skillsheet
Spreadsheets

Technology

Substitution

This spreadsheet shows the values of 5 variables, x, y, z, M and N in cells B1 to B5. To show a fraction in B4, right-click and select **Format Cells**, **Number**, **Fraction** and **Up to 1 digit**.

	A	B	C
1	x=	9	
2	y=	15	
3	z=	-2	
4	M=	1/3	
5	N=	0.1	
6			
7			
8			

1 For each cell given below, enter the correct formula to evaluate the algebraic expression (the first 2 have been done for you). Try to predict the answer before you enter each formula.

a C1, $2x$ Enter: =2*B1
b C2, $\frac{y}{3}$ Enter: =B2/3
c C3, $x - y$
d C4, $5z + y$
e C5, $x + y + z$
f C6, zN
g C7, $\frac{2z}{5}$
h C8, xyz
i C9, M^2
j C10, $N \div M$
k C11, $11 + \frac{z}{4}$
l C12, $yz + x$
m C13, $4(y - z)$
n C14, $z^3 + 2z$

2 Enter the correct formula to evaluate each algebraic expression. Remember to keep the numerator together (the first 2 have been done for you).

a D1, $\frac{x+y}{5}$ Enter: =(B1+B2)/5
b D2, $\frac{3}{x+y}$ Enter: =3/(B1+B2)
c D3, $\frac{5y-6x}{7}$
d D4, $\frac{y}{zN}$
e D5, $\frac{xy}{MN}$
f D6, $\frac{y+x}{y-x}$
g D7, $\frac{xy}{x+y}$
h D8, $\frac{N(5+z)}{M}$
i D9, $\frac{yz}{2} + \frac{x}{N}$
j D10, $\sqrt{x^2 + y^2}$

3 Does $2x^2 = (2x)^2$?

a In cell E1, enter **=2*B1^2** and in cell E2, enter **=(2*B1)^2**.

b Are the values in cells E1 and E2 equal?

4 Does $3z^2 = (3z)^2$?

a In cell E3, enter **=3*B3^2** and in cell E4, enter **=(3*B3)^2**.

b Are the values in cells E3 and E4 equal?

c Does $ax^2 = (ax)^2$ where a is a positive integer?

5 Does $-5y^2 = (-5y)^2$?

a In cell F1, enter **=–5*B2^2** and in cell F2, enter **=(–5*B2)^2**.

b Are the values in cells F1 and F2 equal?

6 Does $-6N^2 = (-6N)^2$?

a In cell F3, enter **=–6*B5^2** and in cell F4, enter **=(–6*B5)^2**.

b Are the values in cells F3 and F4 equal?

c Does $ax^2 = (ax)^2$, where a is a negative integer?

Adding and subtracting terms 1.03

An algebraic expression is made up of **algebraic terms**. For example, $3y - x + 5$ has 3 terms: $3y$, x and 5. **Like terms** are terms whose variables are exactly the same, for example, $3k$ and $7k$, $6ab$ and $2ba$, $9x^2$ and x^2. **Unlike terms** are terms whose variables are different, for example, $2d$ and $5m$, $4xy$ and $9yz$, $4x$ and $2x^2$.

We can add or subtract like terms because their variables represent the same number.

Video
Adding and subtracting terms

Skillsheet
Algebra using diagrams

Interactive
Expression exchange

Example 6

Simplify each algebraic expression.

a $3a + 8a$ **b** $7r - r$ **c** $5xy - 4yx$

Solution

a $3a + 8a = 11a$ '3 lots of any number a' plus '8 lots of the same number a' will give '11 lots of the number a'

b $7r - r = 6r$ '7 lots of any number r' minus '1 lot of the same number r' will leave '6 lots of the number r'

c $5xy - 4yx = 1xy$ xy and yx are the same and can be subtracted

$= xy$ $xy = 1xy$

Algebraic terms are usually written in alphabetical order.

Adding and subtracting terms

Only **like terms** can be added or subtracted.

For example, $2x - 5y$ *cannot* be simplified because x and y represent different numbers.

Example 7

Simplify each expression.

a $2k + 3 + 8k$ **b** $5r + 4s - 7r - 2s$ **c** $11u^2 - 5u + 3u - 7u^2$

Solution

a $2k + 3 + 8k = (2k + 8k) + 3$ Combine the like terms $2k$ and $8k$

$= 10k + 3$ $10k$ and 3 cannot be simplified further

b $5r + 4s - 7r - 2s = 5r - 7r + 4s - 2s$ Group the like terms

$= -2r + 2s$

$= 2s - 2r$ Rewrite with positive term first

c $11u^2 - 5u + 3u - 7u^2 = 11u^2 - 7u^2 - 5u + 3u$

Each + and – sign belongs to the term that follows it.

$= 4u^2 - 2u$

Example 8

Sonja wrote $7a + 3a = 10a^2$. Use the substitution $a = 5$ to test whether her equation is correct or incorrect.

Solution

LHS $= 7a + 3a$ LHS means 'left-hand side' of the equation

$= 7 \times 5 + 3 \times 5$ Using $a = 5$

$= 50$

RHS $= 10a^2$ RHS means 'right-hand side' of the equation

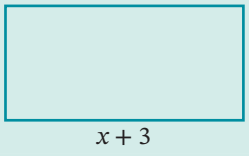
$= 10 \times 5^2$ Using $a = 5$

$= 250$

LHS $\neq$ RHS, so $7a + 3a = 10a^2$ is incorrect.

Example 9

Write a simplified algebraic expression for the perimeter of this rectangle.



Solution

Perimeter $= x + 3 + 2x + x + 3 + 2x$ Adding the lengths of the 4 sides

$= x + 2x + x + 2x + 3 + 3$ Group the like terms

$= 6x + 6$

EXERCISE 1.03 ANSWERS ON P. 562

Adding and subtracting terms U F R

EXAMPLE 6

1 Simplify each algebraic expression.

a	$5y + 6y$	**b**	$12m - 7m$	**c**	$10t + t$	**d**	$15q - q$
e	$5g - 8g$	**f**	$4r^2 + 7r^2$	**g**	$6def - 6def$	**h**	$4xy - 9yx$
i	$5n^2 + 3n$	**j**	$9gh + 12hg$	**k**	$5my^2 + 2m^2y$	**l**	$d - 13d$

EXAMPLE 7

2 Simplify $15 + 8m - 6 + 3m$. Select the correct answer **A**, **B**, **C** or **D**.

A $21 + 11m$ **B** $9 + 5m$ **C** $9 + 11m$ **D** $9 + 5m^2$

3 Simplify each algebraic expression.

a	$2fg + 3fg - 4fg$	**b**	$6uvw - uvw + 4uvw$	**c**	$3t + 3 + 5t$
d	$3x + 2y + 8y$	**e**	$6g + h - 4g$	**f**	$2fg + 3f - 4fg$
g	$7p^2 - 3 - 2p^2$	**h**	$4p - q - 3q$	**i**	$-8r + 6r + 3$
j	$-3a + 4b - 7a$	**k**	$4v + 3 + 2v + 7$	**l**	$2y + 6x + y - 3x$
m	$7p - 8q - p - q$	**n**	$8 - 3w + 7 - 2w$	**o**	$2n - 3 - 12n + 5$
p	$8e - 3f - 2e - 4f$	**q**	$2l^2 + 2l + 3l^2 - 6l$	**r**	$b - 12b^2 + 6b^2 - 15b$
s	$6p^3 - 5p^2 - 4p^3 + p^2$	**t**	$3 - 8k - 5 - 9k$	**u**	$-3 - 5d + 9 - 8d$

4 Simplify $15a^2 + 12a + 48 - 18a$. Select **A**, **B**, **C** or **D**.

A $24a + 48$ **B** $27a^2 + 30a$

C $15a^2 - 6a + 48$ **D** $15a^2 + 30a - 48$

EXAMPLE 8

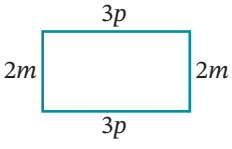
5 Use the substitutions $a = 2$, $b = 3$ and $c = 4$ to test whether each equation is correct or incorrect. R

a	$20c - 12c = 8c$	**b**	$9b + b = 9b^2$	**c**	$7a + 3b = 10 + a + b$
d	$3a^2 + 5a^2 = 8a^2$	**e**	$4ab + 3ab = 7ab$	**f**	$12c - c = 12$

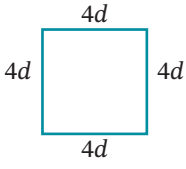
EXAMPLE 9

6 Write a simplified algebraic expression for the perimeter of each shape.

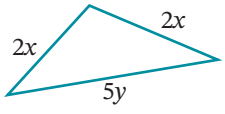
a



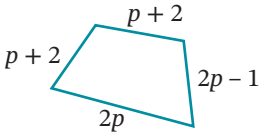
b



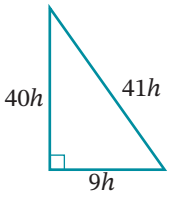
c



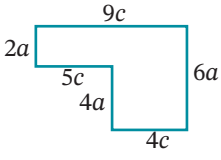
d



e



f



7 Draw a square and write an algebraic expression for its side length so that it has a perimeter of $20x$. R

8 Draw an equilateral triangle and write an algebraic expression for its side lengths so that it has a perimeter of $9d + 6$. R

9 Draw a rectangle and find algebraic expressions for its length and width so that it has a perimeter of $18a - 30$. R

10 Draw a triangle and write algebraic expressions for its side lengths so that it has a perimeter of $18a - 30$. R

1.04 Multiplying and dividing terms

Video Multiplying terms

When writing algebraic expressions, it is not necessary to show the '×' symbol when multiplying terms. For example, $8 \times p \times q$ is abbreviated to $8pq$. When dividing terms, we usually write the expression as a **fraction** rather than use the '÷' symbol. For example, $a \div 3$ is written as $\frac{a}{3}$.

Multiplying and dividing terms

When **multiplying and dividing terms**, multiply or divide the numbers first, then the variables.

Worksheet Perimeter and area

Example 10

Simplify each expression.

a $2 \times 5n \times 4m$ **b** $3d \times 4d$ **c** $2g \times 3h \times (-5f)$

Solution

a $2 \times 5n \times 4m = 2 \times 5 \times 4 \times n \times m$
$= 40mn$

Multiply the numbers and multiply the variables

Remember to write the variables in alphabetical order

b $3d \times 4d = 3 \times 4 \times d \times d$
$= 12d^2$

$d \times d = d^2$

c $2g \times 3h \times (-5f) = 2 \times 3 \times (-5) \times g \times h \times f$
$= -30fgh$

Remember to write the variables in alphabetical order

Foundation Standard Complex

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Example 11

Simplify each expression.

a $\frac{15abc}{3ab}$ **b** $\frac{k^2m}{4km}$ **c** $27cf \div (-3fc)$ **d** $35p^2 \div 15pw$

Solution

a $\frac{15abc}{3ab} = \frac{15^{5}\,a\,b^{1}\,c}{3^{1}\,a\,b^{1}}$

$= 5c$

We can cross out the ab in the **numerator** and **denominator** because $\frac{a}{a} = 1$ and $\frac{b}{b} = 1$

Divide the numbers and divide the variables

b $\frac{k^2m}{4km} = \frac{kkm}{4km}$

$= \frac{1}{4}\,\frac{k\,k\,m^{1}}{k\,m^{1}}$

$= \frac{k}{4}$

$k^2 = k \times k = kk$

Divide the numbers and divide the variables

c $27cf \div (-3fc) = \frac{27cf}{-3fc}$

$= \frac{27}{-3}\,\frac{c\,f^{1}}{c\,f^{1}}$

$= -9$

The variables divide by themselves to give 1

d $35p^2 \div 15pw = \frac{35p^2}{15pw}$

$= \frac{35^{7}}{15^{3}}\,\frac{p^{1}\,p}{p^{1}\,w}$

$= \frac{7p}{3w}$

15 and 35 can both be divided by 5

Example 12

Maddison wrote $7a \times 3a = 21a^2$. Use the substitution $a = 5$ to test whether her equation is correct or incorrect.

Solution

LHS $= 7a \times 3a$

$= 7 \times 5 \times 3 \times 5$

$= 525$

LHS means 'left-hand side' of the equation

Using $a = 5$

RHS $= 21a^2$

$= 21 \times 5^2$

$= 525$

RHS means 'right-hand side' of the equation.

$a = 5$

LHS = RHS, so $7a \times 3a = 21a^2$ is correct.

Video
Dividing terms

1.04

EXERCISE 1.04 ANSWERS ON P. 562

Multiplying and dividing terms U F R

EXAMPLE 10

1 Simplify each expression.

a	$5 \times 8x$	**b**	$5n \times 3$	**c**	$b \times 5a$
d	$7m \times 6t$	**e**	$-4 \times 3w$	**f**	$2k \times (-3p)$
g	$4d \times 3e$	**h**	$6v \times 4w$	**i**	$11q \times 8q$
j	$-5a \times (-4a)$	**k**	$2y \times 3 \times 4b$	**l**	$-6w \times 3w \times 2$
m	$-9c \times (-4h) \times 2k$	**n**	$\frac{1}{3}a \times 24bc$	**o**	$-5d \times (-3e) \times (-2w)$
p	$(5n)^2$	**q**	$(-3r)^2$	**r**	$(at)^2$
s	$-15d \times \frac{2}{3}ad$	**t**	$-4k \times (-\frac{1}{4}ek)$	**u**	$(-2xy)^2$

EXAMPLE 11

2 Simplify each expression.

a	$\frac{12x}{3}$	**b**	$\frac{15y}{y}$	**c**	$\frac{36a}{4a}$
d	$18mn \div 6m$	**e**	$8vw \div 48vw$	**f**	$\frac{64abc}{16b}$
g	$\frac{18g}{-3g}$	**h**	$-25yz \div 5y$	**i**	$\frac{9p}{-45pq}$
j	$\frac{-75xyz}{-25x}$	**k**	$10r^2 \div 40r$	**l**	$-12hk^2 \div 2hk$
m	$24ak^2 \div 8a^2$	**n**	$16mwx \div 12mx$	**o**	$7gh \div 21g$

3 Simplify $\frac{9yx^2}{24xy^2}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{3x^2}{8y^2}$ **B** $\frac{3x^3}{8y^3}$ **C** $\frac{3x}{8y}$ **D** $\frac{3y}{8x}$

4 Simplify each expression.

a	$2a \times 3b \times 5c$	**b**	$4p \times 3 \times 2q$	**c**	$40n^2 \div 5n$
d	$8b^2 \div 40b^2$	**e**	$6h \times 2k \times (-3)$	**f**	$-2 \times 5k \times (-4k)$
g	$\frac{6d}{12}$	**h**	$\frac{7r}{49r}$	**i**	$(-33m) \div 3$
j	$\frac{-9w}{-45w}$	**k**	$8ab \div 2bc$	**l**	$2ef \times (-3fe)$
m	$18ac^2d \div 27acd$	**n**	$-3pq \times (-2q) \times (-4)$	**o**	$\frac{24v^2w}{18v}$

5 Simplify $-5k \times 4km \times (-3m)$. Select **A**, **B**, **C** or **D**.

A $-60k^2m$ **B** $60km$ **C** $-60km^2$ **D** $60k^2m^2$

EXAMPLE 12

6 Use the substitutions $p = 5$, $q = 4$ and $r = 3$ to test whether each equation is correct or incorrect.

a	$3 \times 4p = 12p$	**b**	$q + q = q^2$
c	$2r \times 3r = 6r^2$	**d**	$3q^2 \times 5q^2 = 15q^2$

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e $\frac{12p}{3p} = 4$

f $\frac{7qr}{qr} = 7$

g $3r \times 5q = 15qr$

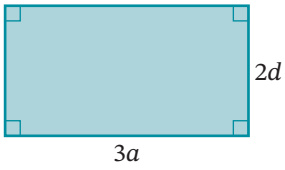
h $p \times q \times q = 2pq$

i $\frac{14r^2}{7r} = 2r$

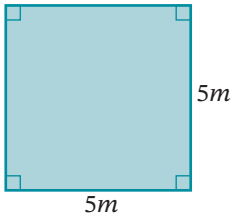
j $\frac{5pq}{10p} = \frac{q}{2}$

7 Write a simplified algebraic expression for the area of each shape.

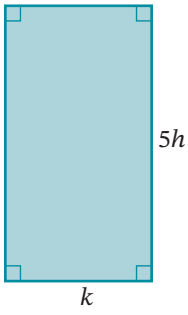
a



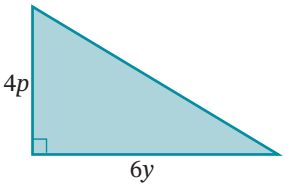
b



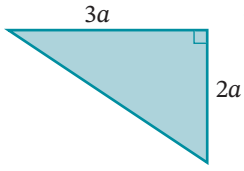
c



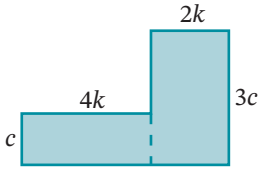
d



e

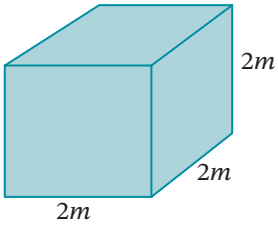


f

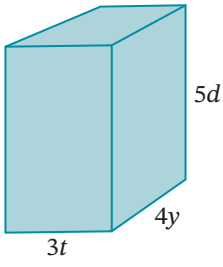


8 Find an algebraic expression for the volume of each rectangular prism.

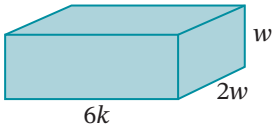
a



b



c



9 Draw a square and write an algebraic expression for its side length if it has an area of $100x^2$.

10 Draw a rectangle and write algebraic expressions for its length and width if it has an area of $36pq$.

Foundation | Standard | Complex

Quiz
Mental skills 1A

Mental skills 1A: Maths without calculators ANSWERS ON P. 563

Multiplying and dividing by 5, 15, 25 and 50

It is easier to multiply or divide a number by 10 than by 5. So whenever we multiply or divide a number by 5, we can double the 5 (to make 10) and then adjust the first number.

1 Study each example.

a **To multiply by 5**, halve the number, then multiply by 10.

$$18 \times 5 = 18 \times \frac{1}{2} \times 10 \quad (\text{or } 9 \times 2 \times 10)$$
$$= 9 \times 10$$
$$= 90$$

b **To multiply by 50**, halve the number, then multiply by 100.

$$26 \times 50 = 26 \times \frac{1}{2} \times 100 \quad (\text{or } 13 \times 2 \times 100)$$
$$= 13 \times 100$$
$$= 1300$$

c **To multiply by 25**, quarter the number, then multiply by 100.

$$44 \times 25 = 44 \times \frac{1}{4} \times 100 \quad (\text{or } 11 \times 4 \times 25)$$
$$= 11 \times 100$$
$$= 1100$$

d **To multiply by 15**, halve the number, then multiply by 30.

$$8 \times 15 = 8 \times \frac{1}{2} \times 30 \quad (\text{or } 4 \times 2 \times 15)$$
$$= 4 \times 30$$
$$= 120$$

e **To divide by 5**, divide by 10 and double the answer. We do this because there are two 5s in every 10.

$$140 \div 5 = 140 \div 10 \times 2$$
$$= 14 \times 2$$
$$= 28$$

f **To divide by 50**, divide by 100 and double the answer. This is because there are two 50s in every 100.

$$400 \div 50 = 400 \div 100 \times 2$$
$$= 4 \times 2$$
$$= 8$$

g **To divide by 25**, divide by 100 and multiply the answer by 4. This is because there are four 25s in every 100.

$$600 \div 25 = 600 \div 100 \times 4$$
$$= 6 \times 4$$
$$= 24$$

h **To divide by 15**, divide by 30 and double the answer. This is because there are two 15s in every 30.

$240 \div 15 = 240 \div 30 \times 2$

$= 8 \times 2$

$= 16$

2 Now evaluate each expression.

a	32×5	**b**	14×5	**c**	48×5	**d**	18×50
e	52×50	**f**	36×25	**g**	28×5	**h**	12×25
i	12×15	**j**	22×35	**k**	$90 \div 5$	**l**	$170 \div 5$
m	$230 \div 5$	**n**	$1300 \div 50$	**o**	$900 \div 50$	**p**	$300 \div 25$
q	$1000 \div 25$	**r**	$360 \div 45$	**s**	$210 \div 15$	**t**	$360 \div 15$

Adding and subtracting algebraic fractions

1.05

Adding and subtracting fractions

To **add or subtract fractions**, convert them (if needed) so that they have the same denominator, then simply add or subtract the numerators.

Example 13

Simplify each expression.

a $\frac{7r}{10} - \frac{3r}{10}$ **b** $\frac{k}{3} + \frac{k}{2}$ **c** $\frac{3h}{4} + \frac{2h}{5}$

d $\frac{5m}{6} - \frac{2m}{3}$ **e** $\frac{3}{4} + \frac{7a}{10}$ **f** $\frac{5x}{8} - \frac{2w}{3}$

Solution

a $\frac{7r}{10} - \frac{3r}{10} = \frac{4r}{10}$

$= \frac{2r}{5}$

b $\frac{k}{3} + \frac{k}{2} = \frac{2 \times k}{2 \times 3} + \frac{3 \times k}{3 \times 2}$ Common denominator $= 2 \times 3 = 6$

$= \frac{2k}{6} + \frac{3k}{6}$

$= \frac{5k}{6}$

Puzzle
Algebraic fractions

Video
Adding and subtracting algebraic fractions

Presentation
Adding and subtracting fractions

c $\frac{3h}{4} + \frac{2h}{5} = \frac{5 \times 3h}{5 \times 4} + \frac{4 \times 2h}{4 \times 5}$ Common denominator $= 5 \times 4 = 20$

$= \frac{15h}{20} + \frac{8h}{20}$

$= \frac{23h}{20}$

d $\frac{5m}{6} - \frac{2m}{3} = \frac{15m}{18} - \frac{12m}{18}$

$= \frac{3m}{18}$

$= \frac{m}{6}$

or $\frac{5m}{6} - \frac{2m}{3} = \frac{5m}{6} - \frac{4m}{6}$ The lowest common denominator is 6

$= \frac{m}{6}$

e $\frac{3}{4} + \frac{7a}{10} = \frac{15}{20} + \frac{14a}{20}$

$= \frac{15+14a}{20}$

f $\frac{5x}{8} - \frac{2w}{3} = \frac{15x}{24} - \frac{16w}{24}$

$= \frac{15x - 16w}{24}$

EXERCISE 1.05 ANSWERS ON P. 563

Adding and subtracting algebraic fractions U F

EXAMPLE 13

1 Simplify each expression.

a $\frac{w}{4} + \frac{2w}{4}$ **b** $\frac{4k}{8} + \frac{7k}{8}$ **c** $\frac{7m}{10} - \frac{2m}{10}$ **d** $\frac{x}{3} + \frac{2x}{3}$

e $\frac{4}{q} + \frac{5}{q}$ **f** $\frac{5}{3d} - \frac{2}{3d}$ **g** $\frac{4r}{w} + \frac{r}{w}$ **h** $\frac{5}{c} - \frac{2}{c}$

i $\frac{4t}{3} + \frac{s}{3}$ **j** $\frac{11y}{2h} - \frac{7y}{2h}$ **k** $\frac{6}{a} - \frac{5}{a}$ **l** $\frac{8}{5e} + \frac{7}{5e}$

m $\frac{p}{z} + \frac{3}{z}$ **n** $\frac{5u}{8g} - \frac{3u}{8g}$ **o** $\frac{4}{9f} - \frac{1}{9f}$ **p** $\frac{7e}{10} - \frac{3e}{10}$

q $\frac{5a}{6} + \frac{5a}{6}$ **r** $\frac{5}{2d} + \frac{3}{2d}$ **s** $\frac{7}{5k} + \frac{13}{5k}$ **t** $\frac{12}{4a} - \frac{8}{4a}$

2 Simplify the expression $\frac{9}{4k} - \frac{3}{4k}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{3}{k}$ **B** $\frac{3}{4k}$ **C** $\frac{3}{2k}$ **D** $\frac{6}{k}$

3 Simplify each expression.

a $\frac{x}{3} + \frac{x}{4}$ **b** $\frac{s}{3} - \frac{s}{7}$ **c** $\frac{h}{5} + \frac{h}{3}$ **d** $\frac{m}{7} - \frac{m}{2}$

e $\frac{w}{4} + \frac{w}{5}$ **f** $\frac{5t}{4} - \frac{2t}{5}$ **g** $\frac{2p}{5} + \frac{p}{3}$ **h** $\frac{5r}{2} - \frac{5r}{3}$

i $\frac{5y}{6} + \frac{2y}{3}$ **j** $\frac{3m}{4} - \frac{5m}{12}$ **k** $\frac{5a}{9} - \frac{a}{6}$ **l** $\frac{3y}{10} + \frac{7y}{15}$

Foundation Standard Complex

m $\frac{3c}{2}-\frac{c}{5}$ **n** $\frac{2d}{11}-\frac{r}{3}$ **o** $\frac{3h}{5}+\frac{2d}{3}$ **p** $\frac{5}{6}+\frac{4w}{5}$

q $\frac{2a}{7}+\frac{3}{4}$ **r** $\frac{4e}{7}-\frac{2e}{5}$ **s** $\frac{3m}{2}-\frac{5n}{11}$ **t** $\frac{2k}{5}+\frac{m}{6}$

4 Simplify the expression $\frac{5m}{12}-\frac{3m}{8}$. Select **A**, **B**, **C** or **D**.

A $\frac{2m}{4}$ **B** $\frac{m}{24}$ **C** $\frac{m}{2}$ **D** $\frac{3m}{20}$

5 Simplify the expression $\frac{4h}{9}+\frac{2d}{5}$. Select **A**, **B**, **C** or **D**.

A $\frac{36h+10d}{45}$ **B** $\frac{4h+2d}{14}$ **C** $\frac{20h+18d}{45}$ **D** $\frac{9h+18d}{14}$

Multiplying and dividing algebraic fractions

1.06

Multiplying and dividing algebraic fractions

- To **multiply fractions**, cancel any common factors, then multiply the numerators and denominators separately
- To **divide by a fraction** $\frac{a}{b}$, multiply by its **reciprocal** $\frac{b}{a}$

Puzzles
Algebraic fractions puzzle

Upside-down fractions

Example 14

Simplify each product.

a $\frac{3}{d}\times\frac{4}{c}$ **b** $\frac{4}{k}\times\frac{3k}{16}$ **c** $\frac{5}{2r}\times\frac{8r}{11}$

Solution

a $\frac{3}{d}\times\frac{4}{c}=\frac{3\times4}{d\times c}$

$=\frac{12}{dc}$

b $\frac{4}{k}\times\frac{3k}{16}=\frac{{}^{1}\cancel{4}}{{}_{1}\cancel{k}}\times\frac{3\cancel{k}}{\cancel{16}_{4}}$

$=\frac{3}{4}$

c $\frac{5}{2r}\times\frac{8r}{11}=\frac{5}{{}_{1}\cancel{2r}}\times\frac{\cancel{8r}^{4}}{11}$

$=\frac{20}{11}$

$=1\frac{9}{11}$

Example 15

Simplify each quotient.

a $\frac{3}{h}\div\frac{4}{k}$ **b** $\frac{xy}{5}\div\frac{3x}{25}$

Foundation Standard Complex

Solution

a $\frac{3}{h} \div \frac{4}{k} = \frac{3}{h} \times \frac{k}{4}$
$= \frac{3k}{4h}$

The reciprocal of $\frac{4}{k}$ is $\frac{k}{4}$.

b $\frac{xy}{5} \div \frac{3x}{25} = \frac{{}^{1}\cancel{x}y}{\cancel{5}_{1}} \times \frac{25^{5}}{3\cancel{x}^{1}}$
$= \frac{5y}{3}$

EXERCISE 1.06 ANSWERS ON P. 563

Multiplying and dividing algebraic fractions U F

EXAMPLE 14

1 Simplify the product $\frac{3x}{5} \times \frac{w}{6}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{xw}{10}$ **B** $\frac{3xw}{11}$ **C** $\frac{3xw}{30}$ **D** $\frac{18x+5w}{30}$

2 Simplify each product.

a $\frac{w}{3} \times \frac{1}{2}$ **b** $\frac{s}{5} \times \frac{t}{4}$ **c** $\frac{3}{h} \times \frac{5}{6k}$ **d** $\frac{4}{m} \times \frac{3}{n}$

e $\frac{l}{3} \times \frac{5}{f}$ **f** $\frac{1}{v} \times \frac{2}{3v}$ **g** $\frac{2}{x} \times \frac{3}{x}$ **h** $\frac{a}{b} \times \frac{c}{d}$

i $\frac{d}{e} \times \frac{e}{g}$ **j** $\frac{4ad}{9} \times \frac{d}{16a}$ **k** $\frac{5p}{4} \times \frac{8}{15p}$ **l** $\frac{4}{ak} \times \frac{3a}{5k}$

m $\frac{2}{3} \times \frac{9u}{10}$ **n** $\frac{u}{3} \times \frac{3}{u}$ **o** $\frac{3z}{r} \times \frac{2r}{9dz}$

EXAMPLE 15

3 Simplify the quotient $\frac{3}{4m} \div \frac{9n}{8m}$. Select **A**, **B**, **C** or **D**.

A $\frac{3n}{2}$ **B** $\frac{32m^2}{27n}$ **C** $\frac{36m}{24n}$ **D** $\frac{2}{3n}$

4 Simplify each quotient.

a $\frac{r}{2} \div \frac{r}{5}$ **b** $\frac{m}{6} \div \frac{n}{3}$ **c** $\frac{h}{2} \div \frac{h}{8}$ **d** $\frac{q}{4} \div \frac{3}{4}$

e $\frac{3y}{5} \div \frac{2y}{d}$ **f** $\frac{4t}{9} \div \frac{3t}{5}$ **g** $\frac{3}{e} \div \frac{5}{e}$ **h** $\frac{3}{2a} \div \frac{b}{6a}$

i $\frac{5m}{n} \div \frac{2m}{3n}$ **j** $\frac{h}{k} \div \frac{k}{h}$ **k** $\frac{8w}{3x} \div \frac{2w}{9x}$ **l** $\frac{3s}{4} \div \frac{6s}{11}$

m $\frac{t}{3} \div \frac{3t}{5u}$ **n** $\frac{3e}{7g} \div \frac{e}{14g}$ **o** $\frac{xh}{5} \div \frac{3h}{15}$

5 Simplify each expression.

a $\frac{3p}{2} \div \frac{p^2}{4}$ **b** $\frac{2w}{7} \div \frac{5w}{6}$ **c** $\frac{8y}{5} \times \frac{3}{32y^2}$ **d** $\frac{5}{2g} \div \frac{2}{g}$

e $\frac{xy}{z} \times \frac{5}{y}$ **f** $\frac{a}{b} \times \frac{c}{b} \times \frac{b}{3a}$ **g** $5 \div \frac{3}{5b}$ **h** $\frac{4t}{9} \div 3$

i $\frac{5mn}{2d} \times \frac{4d}{n} \times \frac{1}{15mn}$ **j** $5p \div \frac{7}{10pt}$ **k** $\frac{s}{3} \div \frac{5s}{2} \times \frac{3s}{7}$ **l** $\frac{7}{h} \times \frac{1}{42h}$

m $\frac{5ty}{k} \times \frac{5ky}{t}$ **n** $\frac{6}{r} \times \frac{5r}{9} \div \frac{15}{yh}$ **o** $\frac{6cf}{q} \div \frac{5}{qf} \div \frac{10c}{a}$

6 Simplify the expression $\frac{4ab}{5} \div \frac{a}{2b} \div \frac{8b}{15}$. Select **A**, **B**, **C** or **D**.

A 3 **B** $3b$ **C** $\frac{3b}{5}$ **D** $\frac{3b}{a}$

Foundation Standard Complex

Expanding expressions

1.07

When an algebraic expression such as $5(x + 2)$ is simplified by removing the grouping symbols (brackets) by multiplying, we call it **expanding** the expression.

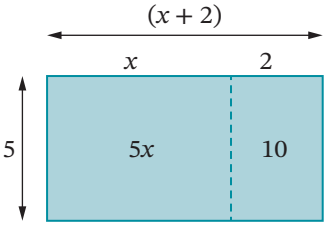
For example, $5(x + 2)$ means '5 times $(x + 2)$', so if we expand 'the long way':

$$5(x + 2) = (x + 2) + (x + 2) + (x + 2) + (x + 2) + (x + 2)$$
$$= x + x + x + x + x + 2 + 2 + 2 + 2 + 2$$
$$= 5x + 10$$

We can expand $5(x + 2)$ more quickly by noticing that:

$$5(x + 2) = (5 \times x) + (5 \times 2)$$
$$= 5x + 10$$

Each term inside the grouping symbols $(x + 2)$ is multiplied by the term outside the grouping symbols (5). This is called the **distributive law** because the 5 is distributed across the x and the 2 inside the grouping symbols. This can also be demonstrated using an area diagram.



Area of rectangle $= 5(x + 2) = 5x + 10$

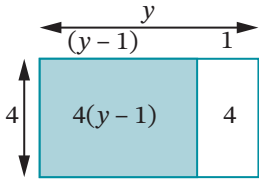
We can expand $4(y - 1)$ in similar ways.

This can also be demonstrated using an area diagram.

$4(y - 1) = (y - 1) + (y - 1) + (y - 1) + (y - 1)$ or more quickly:

$$= y + y + y + y - 1 - 1 - 1 - 1$$
$$= 4y - 4$$

$$4(y - 1) = (4 \times y) - (4 \times 1)$$
$$= 4y - 4$$



Area of left rectangle (shaded) $= 4(y - 1)$ (area of whole rectangle – area of right rectangle)

$$= 4y - 4$$

The distributive law for expanding an expression

Multiply each term inside the brackets by the term outside

$$a(b + c) = ab + ac$$
$$a(b - c) = ab - ac$$

Skillsheet
Algebra using diagrams

Videos
Expanding expressions 1
Expanding expressions 2

Worksheet
Algebra review

Interactive
Area model Algebra

Example 16

Expand each expression.

a $4(5 + d)$ **b** $2w(3w + 10)$

Solution

a $4(5 + d) = 4 \times 5 + 4 \times d$
$= 20 + 4d$

multiply each term inside the brackets by the term outside

$a(b + c) = ab + ac$

b $2w(3w + 10) = 2w \times 3w + 2w \times 10$
$= 6w^2 + 20w$

Example 17

Expand each expression.

a $3g(h - 2)$ **b** $r(10 - 4r)$

Solution

a $3g(h - 2) = 3g \times h - 3g \times 2$
$= 3gh - 6g$

multiply each term inside the brackets by the term outside

$a(b - c) = ab - ac$

b $r(10 - 4r) = r \times 10 - r \times 4r$
$= 10r - 4r^2$

Example 18

Expand each expression.

a $-(y + 4)$ **b** $-3(2m - 7)$

Solution

a $-(y + 4) = -1(y + 4)$
$= -y + (-4)$
$= -y - 4$

$-$ ( ) is the same as $-1 \times$ ( )

b $-3(2m - 7) = -3 \times 2m - (-3) \times 7$
$= -6m - (-21)$
$= -6m + 21$

Example 19

Expand and simplify by collecting like terms.

a $4(3p + 2) - 5p$ **b** $5(2e - 3) - 4(1 - 5e)$ **c** $7(n - 3) + n(n - 1)$

Solution

a $4(3p + 2) - 5p = 4 \times 3p + 4 \times 2 - 5p$ — Expanding

$= 12p + 8 - 5p$ — Collecting like terms to simplify

$= 12p - 5p + 8$

$= 7p + 8$

b $5(2e - 3) - 4(1 - 5e) = 5 \times 2e - 5 \times 3 - 4 \times 1 - (-4) \times 5e$ — Expanding

$= 10e - 15 - 4 - (-20e)$ — Collecting like terms to simplify

$= 10e - 19 + 20e$

$= 30e - 19$

c $7(n - 3) + n(n - 1) = 7n - 21 + n^2 - n$ — It's conventional to place n^2 first

$= n^2 + 7n - n - 21$ — Collecting like terms to simplify

$= n^2 + 6n - 21$

EXERCISE 1.07 ANSWERS ON P. 563

Expanding expressions U F

1 Use an area diagram to expand each expression.

a $3(x + 5)$ **b** $8(e + 9)$ **c** $2n(n + 3)$

2 Expand each expression. EXAMPLE 16

a $4(h + 6)$ **b** $5(3 + t)$ **c** $3(2b + 7)$

d $5(8 + 3x)$ **e** $x(x + 9)$ **f** $p(5 + p)$

g $3r(2r + 1)$ **h** $4y(1 + 3y)$ **i** $10e(2e + 4f)$

3 Use an area diagram to expand each expression.

a $10(b - 2)$ **b** $6(2y - 3)$ **c** $4t(5t - 4)$

4 Expand each expression. EXAMPLE 17

a $3(t - 2)$ **b** $7(5 - d)$ **c** $8(8g - 3)$

d $w(w - 7)$ **e** $a(a - 1)$ **f** $6h(3h - 1)$

g $4x(3x - 1)$ **h** $5x(2x - 4y)$ **i** $3a(4b - 7a)$

5 Expand $-5u(8 + 2u)$. Select the correct answer **A**, **B**, **C** or **D**.

A $3u - 3u^2$ **B** $-40u + 10u^2$ **C** $-40u - 10u^2$ **D** $-40u - 3u^2$

6 Use the substitution $x = 5$ to test whether each equation is correct or incorrect.

a $3(x + 3) = 3x + 6$ **b** $x(4 - 2x) = 4x - 2x^2$

7 Expand each expression. EXAMPLE 18

a $-(a - 5)$ **b** $-(a + 5)$ **c** $-2(x + 6)$

d $-2(x - 6)$ **e** $-(11 + w)$ **f** $-(11 - w)$

g $-y(y + 9)$ **h** $-x(8 - z)$ **i** $-4t(t - 8)$

Foundation Standard Complex

8 Expand $-k(3k - 7)$. Select **A**, **B**, **C** or **D**.

A $-2k^2 - 8k$ **B** $-3k^2 + 7k$ **C** $-3k^2 - 7k$ **D** $-3k + 7k^2$

9 Expand $3(2 - 7y) + 5y$. Select **A**, **B**, **C** or **D**.

A $6 - 26y$ **B** $6 + 16y$ **C** $6 - 5y$ **D** $6 - 16y$

EXAMPLE 19

10 Expand and simplify each expression.

a $5(3m + 2) + 4m$ **b** $3(1 - 5e) + 6e$

c $4w - 2(5 + 2w)$ **d** $8 - 5(2x - 7)$

e $5(2a + 3) + 4(a + 7)$ **f** $3(2d + 3) - 5(d + 4)$

g $7(3g - 1) + 4(2g - 3)$ **h** $4(3 - 4w) - 2(w - 5)$

i $5(6c - 3) - 3(4 - 3c)$ **j** $t(t + 4) + 3(t + 4)$

k $4(3 + h) + h(7 - 2h)$ **l** $6(2e - 1) - (5 - 3e)$

m $3x(2x + 5) + 4(2x + 5)$ **n** $v(2v + 3) - 6(v + 1)$

o $3(1 - 2w) - w(2 - w)$ **p** $2y(3y - 7) - 5(3y - 7)$

q $4(3x + 2y - 1) - (x - 1)$ **r** $4x(3x - 2y + 1) - (x + 1)$

11 Expand $3(1 - 4k) - 2(3k + 7)$. Select **A**, **B**, **C** or **D**.

A $17 - 18k$ **B** $11 - 6k$ **C** $-11 - 18k$ **D** $-17 + 6k$

1.08 Factorising expressions

Skillsheets HCF by factor trees

Factorising using diagrams

Factors

A **factor** (or **divisor**) is a number or algebraic expression that **divides evenly** into a larger number or algebraic expression. For example, 5 is a factor of 10 because $10 \div 5 = 2$, and $5x$ is a factor of $10xy$ because $10xy \div 5x = 2y$.

- The factors of 10 are 1, 2, 5, 10.
- *Some* of the factors of $10xy$ are 1, 5, $2x$, $5x$, $10x$, $10y$, xy, $2xy$ and $10xy$.

$3a$ is a factor of both $15a^2$ and $30a$ because it divides into both terms. We can say that $3a$ is a **common factor** of $15a^2$ and $30a$.

video HCF of algebraic terms

The highest common factor (HCF)

The **highest common factor (HCF)** or **greatest common divisor (GCD)** of 2 or more terms is the largest term that is a factor of all of the terms. To find the HCF of algebraic terms:

- find the HCF of the numbers
- find the HCF of the variables
- multiply them together

Puzzle Factorising puzzle

Foundation Standard Complex

9780170479370

Example 20

Find the highest common factor (HCF) of each pair of terms.

a $20x^2$ and $15xy$ **b** 16 and $12b$

Solution

Find the HCF of the numbers and the HCF of the variables. Their product is the HCF of the expression.

a The HCF of 20 and 15 is 5.

The HCF of x^2 and xy is x, since it is the 'largest' common part of x^2 and xy.

$\therefore$ The HCF of $20x^2$ and $15xy = 5 \times x = 5x$

b The HCF of 16 and 12 is 4.

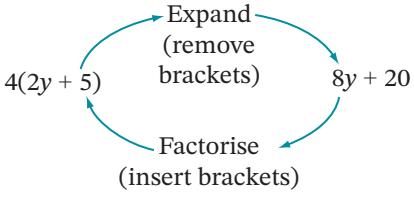
There is **no** HCF of the variables because b is in $12b$ but not in 16.

$\therefore$ The HCF of 16 and $12b$ is 4.

Factorising algebraic expressions

When $4(2y + 5)$ is **expanded**, the answer is $8y + 20$.

Factorising is the reverse of expanding. Factorisation breaks an expression into factors. To factorise $8y + 20$, we take out the highest common factor and **insert** brackets. The result is $4(2y + 5)$. The factors are 4 and $(2y + 5)$.



Example 21

Complete each factorisation by inserting the highest common factor.

a $5a + 15 = ___ (a + 3)$ **b** $x^2 - 7x = ___ (x - 7)$

c $y(y + 1) - 3(y + 1) = ____ (y - 3)$

Solution

a $5a + 15 = 5(a + 3)$ The HCF is 5

You can always check your answer by expanding it: $5(a + 3) = 5a + 15$

b $x^2 - 7x = x(x - 7)$ The HCF is x

c $y(y + 1) - 3(y + 1) = (y + 1)(y - 3)$ The HCF is $(y + 1)$

Factorising an expression

- Find the HCF of the terms and write it outside the brackets
- Divide each term by the HCF and write the answers inside the brackets

$$ab + ac = a(b + c)$$

$$ab - ac = a(b - c)$$

- To check that the factorised answer is correct, expand it

Video
Factorising expressions

Example 22

Factorise each expression.

a $8y + 16$ **b** $25b^2 - 20ab$ **c** $v(4 + w) + 2(4 + w)$

Solution

a The HCF of $8y$ and 16 is 8.

$\therefore 8y + 16 = 8 \times y + 8 \times 2$ Rewrite the expression using the HCF 8

$= 8(y + 2)$ Write the HCF at the front of the brackets

b The HCF of $25b^2$ and $20ab$ is $5b$.

$\therefore 25b^2 - 20ab = 5b \times 5b - 5b \times 4a$ Rewrite the expression using the HCF $5b$

$= 5b(5b - 4a)$ Write the HCF at the front of the brackets

c The HCF of $v(4 + w) + 2(4 + w)$ is $(4 + w)$.

$\therefore v(4 + w) + 2(4 + w) = (4 + w) \times v + (4 + w) \times 2$

$= (4 + w)(v + 2)$

Factorising with negative terms

Example 23

Factorise each expression.

a $-x^2 + 3x$ **b** $-a - ab$

Solution

When factorising expressions that begin with a **negative** term, we use the 'negative' HCF.

a The highest 'negative' common factor of $-x^2$ and $3x$ is $-x$.

$\therefore -x^2 + 3x = (-x) \times x + (-x) \times (-3)$

$= (-x)[x + (-3)]$ $(-x) \times (-3) = +3x$

$= -x(x - 3)$ Check your answer by expanding.

b The highest 'negative' common factor of $-a$ and ab is $-a$.

$\therefore -a - ab = -a \times 1 + (-a) \times b$

$= -a(1 + b)$ $(-a) \times b = -ab$

Factorising with negative terms

- Find the 'negative' HCF of the terms and write it outside the brackets
- Divide each term by the HCF and write the answers inside the brackets

$$-ab - ac = -a(b + c)$$

$$-ab + ac = -a(b - c)$$

- To check that the factorised answer is correct, expand it

EXERCISE 1.08 ANSWERS ON P. 563

Factorising expressions U F

EXAMPLE 20

1 Find the HCF of each pair of terms.

a $12, 6y$ **b** $8a^2, 24$ **c** $20b, 15$

d $6, 9p^2$ **e** $24mn, m$ **f** c, c^2

g $18pq, 12p^2$ **h** $24w^2, 16w$ **i** $3(x - 5), x(x - 5)$

EXAMPLE 21

2 Copy and complete each factorisation.

a $4a + 12 = ___(a + 3)$ **b** $a^2 + 10a = ___(a + 10)$

c $10b - 40 = ___(b - 4)$ **d** $18p^2 + 24 = 6(___ + ___)$

e $9m^2 + 3m = 3m(___ + ___)$ **f** $20d - 30d^2 = 10d(___ - ___)$

g $6x^2y - 8xy = 2xy(___ - ___)$ **h** $x(x - 2) + 5(x - 2) = (x - 2)(___ + ___)$

i $n(3 + n) - 3(3 + n) = (3 + n)(___ - ___)$

j $v(1 - w) + (1 - w) = (___ + ___)(1 - w)$

EXAMPLE 22

3 Factorise each expression.

a $3f + 6$ **b** $4m - 28$ **c** $9n + 27$

d $16 + 24t$ **e** $12q - 18$ **f** $24x + 30$

g $20g^2 - 64$ **h** $xy + y$ **i** $mn^2 - 3n$

j $2y - y^2$ **k** $16r^2 - 12r$ **l** $6t^2 + 27t$

4 Factorise each expression.

a $12x^2y - 16x$ **b** $18p^2 + 16pr$ **c** $4m^2n - 4mn^2$

d $14abc + 21bc$ **e** $28vw - 21v^2w^2$ **f** $45rt + 54r^2t$

5 Factorise completely $24xy^2 - 16xyw$. Select the correct answer **A**, **B**, **C** or **D**.

A $8xy(3y - 2w)$ **B** $2x(12y^2 - 8yw)$

C $4xy(6y - 4w)$ **D** $8xy^2(3 - 2w)$

6 Factorise each expression.

a $a(a - 3) + 6(a - 3)$ **b** $t(8 + t) - 3(8 + t)$

c $b(b + 5) - 2(b + 5)$ **d** $x(2 - y) - 6(2 - y)$

e $5(a - 7) + b(a - 7)$ **f** $s(a + 3) + 4(a + 3)$

g $3p(g + 5) - 2(g + 5)$ **h** $5(2 - 3m) + 4n(2 - 3m)$

i $r(8 + r) + (8 + r)$ **j** $(y - 6) - y(y - 6)$

Foundation Standard Complex

EXAMPLE 23

7 Factorise each expression using the 'negative' HCF.

a	$-4q - 8$	**b**	$-9u + 18$	**c**	$-3g + 6$
d	$-18a + 12$	**e**	$-n - 1$	**f**	$-n + 1$
g	$-y^2 - 9y$	**h**	$-6t + 10t^2$	**i**	$-3a^2 - 6ab$
j	$-ap + aq$	**k**	$-20e^2 - 22e$	**l**	$-9m + 3m^2$

8 Factorise $-10kr + 4rn$. Select **A**, **B**, **C** or **D**.

A	$-2r(5k - 4n)$	**B**	$-2r(5k - 2n)$
C	$-5r(2k - 2n)$	**D**	$-2r(5k + 2n)$

Quiz
Mental skills 1B

Mental skills 1B: Maths without calculators ANSWERS ON P. 564

Multiplying by 9, 11, 99 and 101

We can use expansion when multiplying by a number near 10 or near 100.

1 Study each example.

a
$$\begin{aligned} 25 \times 11 &= 25 \times (10 + 1) \\ &= 25 \times 10 + 25 \times 1 \\ &= 250 + 25 \\ &= 275 \end{aligned}$$

b
$$\begin{aligned} 14 \times 9 &= 14 \times (10 - 1) \\ &= 14 \times 10 - 14 \times 1 \\ &= 140 - 14 \\ &= 126 \end{aligned}$$

c
$$\begin{aligned} 32 \times 12 &= 32 \times (10 + 2) \\ &= 32 \times 10 + 32 \times 2 \\ &= 320 + 64 \\ &= 384 \end{aligned}$$

d
$$\begin{aligned} 7 \times 99 &= 7 \times (100 - 1) \\ &= 7 \times 100 - 7 \times 1 \\ &= 700 - 7 \\ &= 693 \end{aligned}$$

e
$$\begin{aligned} 27 \times 101 &= 27 \times (100 + 1) \\ &= 27 \times 100 + 27 \times 1 \\ &= 2700 + 27 \\ &= 2727 \end{aligned}$$

f
$$\begin{aligned} 18 \times 8 &= 18 \times (10 - 2) \\ &= 18 \times 10 - 18 \times 2 \\ &= 180 - 36 \\ &= 144 \end{aligned}$$

2 Now evaluate each product.

a	16×11	**b**	33×11	**c**	29×9	**d**	45×9
e	62×11	**f**	7×101	**g**	18×101	**h**	36×99
i	19×8	**j**	45×12	**k**	21×102	**l**	6×98
m	32×9	**n**	7×99	**o**	39×101	**p**	71×12

Expanding binomial products

1.09

$(x + 5)$ and $(x - 1)$ are called **binomial expressions** because they each have exactly 2 terms.

$(x + 5)(x - 1)$ is called a **binomial product** because it is a product of 2 binomial expressions.

binomial = '2 terms'

Worksheets
Area diagrams
Binomial products

Puzzle
Expanding brackets

Technology
Expanding binomials

Spreadsheet
Expanding binomials

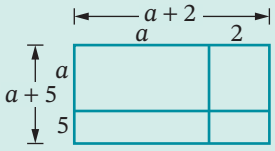
Example 24

Expand each binomial product using an area diagram.

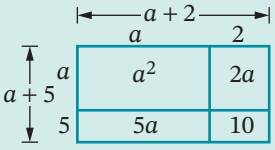
a $(a + 2)(a + 5)$ **b** $(n + 4)(n - 3)$

Solution

a Draw an area diagram (rectangle) with length $(a + 2)$ and width $(a + 5)$ and divide the diagram into 4 smaller rectangles.



Find the area of each smaller rectangle.

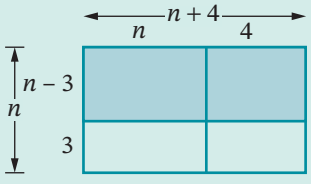


Expand $(a + 2)(a + 5)$ by adding the areas of the 4 rectangles.

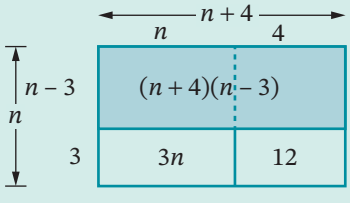
$(a + 2)(a + 5) = a^2 + 2a + 5a + 10$ Expanding

$= a^2 + 7a + 10$ Simplifying by collecting like terms

b Draw an area diagram with length $(n + 4)$ and width $(n - 3)$, then increase the width to n by adding 3. This creates 4 rectangles.



The 2 shaded upper rectangles together give the product $(n + 4)(n - 3)$. To find its area, we subtract the areas of the 2 lower rectangles from the area of the whole rectangle.



$(n + 4)(n - 3) = n(n + 4) - 3n - 12$ Whole rectangle − 2 lower rectangles

$= n^2 + 4n - 3n - 12$ Expanding

$= n^2 + n - 12$ Simplifying by collecting like terms

When expanding a binomial product algebraically, each term in the first binomial is multiplied by each term in the second binomial to give **4 terms**, which are collected and simplified.

Video
Expanding binomial products 1

Example 25

Expand each binomial product.

a $(x + 5)(x + 9)$ **b** $(k + 3)(k - 7)$

c $(7 - m)(4 + m)$ **d** $(a - 6)^2$

Solution

a $(x + 5)(x + 9) = x(x + 9) + 5(x + 9)$ Each term in $(x + 5)$ is multiplied by $(x + 9)$

$= x^2 + 9x + 5x + 45$ Expanding to make 4 terms

$= x^2 + 14x + 45$ Adding $9x$ and $5x$

b $(k + 3)(k - 7) = k(k - 7) + 3(k - 7)$ Each term in $(k + 3)$ is multiplied by $(k - 7)$

$= k^2 - 7k + 3k - 21$ Expanding to make 4 terms

$= k^2 - 4k - 21$ Adding $-7k$ and $3k$

c $(7 - m)(4 + m) = 7(4 + m) - m(4 + m)$ Each term in $(7 - m)$ is multiplied by $(4 + m)$

$= 28 + 7m - 4m - m^2$ Expanding to make 4 terms

$= 28 + 3m - m^2$ Adding $7m$ and $-4m$

d $(a - 6)^2 = (a - 6)(a - 6)$

$= a(a - 6) - 6(a - 6)$ Each term in $(a - 6)$ is multiplied by $(a - 6)$

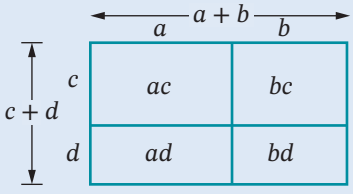
$= a^2 - 6a - 6a + 36$ Expanding to make 4 terms

$= a^2 - 12a + 36$ Adding $-6a$ and $-6a$

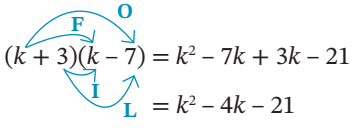
The distributive law for expanding a binomial product

Multiply each term in the first binomial by each term in the second binomial.

$$(a + b)(c + d) = ac + ad + bc + bd$$



One way of remembering which pairs of terms to multiply together in a binomial product is called the **FOIL method**, as shown below.



- **F** means multiply the **first** terms: $k \times k = k^2$
- **O** means multiply the **outside** terms: $k \times (-7) = -7k$
- **I** means multiply the **inside** terms: $3 \times k = 3k$
- **L** means multiply the **last** terms: $3 \times (-7) = -21$

Example 26

Expand each binomial product.

a $(x - 6)(4x + 2)$ **b** $(3t - 1)(2t - 5)$ **c** $(2x + 5)(3y - 4)$

Solution

a $(x - 6)(4x + 2) = x(4x + 2) - 6(4x + 2)$

$= 4x^2 + 2x - 24x - 12$ Expanding

$= 4x^2 - 22x - 12$ Simplifying

b $(3t - 1)(2t - 5) = 3t(2t - 5) - 1(2t - 5)$

$= 6t^2 - 15t - 2t + 5$ Expanding

$= 6t^2 - 17t + 5$ Simplifying

c $(2x + 5)(3y - 4) = 2x(3y - 4) + 5(3y - 4)$

$= 6xy - 8x + 15y - 20$ Expanding

Videos
Expanding binomial products 1
Expanding binomial products 2

1.09

EXAMPLE 24

EXERCISE 1.09 ANSWERS ON P. 564

Expanding binomial products UFRC

1 Expand each binomial product by copying and completing the area diagram.

a $(x + 10)(x + 8)$

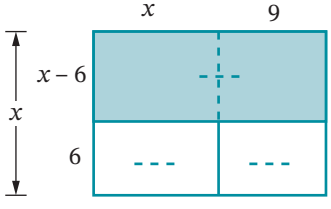
	x	10
x	---	---
8	---	---

b $(4h + 1)(3h + 7)$

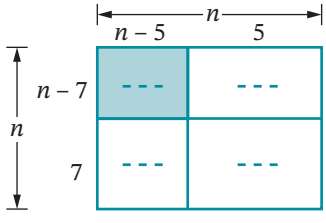
	$4h$	1
$3h$	---	---
7	---	---

2 Expand each binomial product by copying and completing the area diagram. (Find the shaded area.)

a $(x + 9)(x - 6)$



b $(n - 5)(n - 7)$



3 Draw an area diagram for each binomial product and use it to expand the product.

a $(2y + 5)(y + 3)$ **b** $(w - 5)(w + 3)$ **c** $(p - 3)(p - 5)$

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EXAMPLE 25

4 Expand each binomial product algebraically.

a	$(n + 6)(n + 3)$	**b**	$(t + 6)(t + 5)$	**c**	$(p + 10)(p - 10)$
d	$(x - 8)(x + 3)$	**e**	$(b - 2)(9 + b)$	**f**	$(u - 8)(u - 7)$
g	$(15 - r)(r + 1)$	**h**	$(a - 10)(a - 9)$	**i**	$(5 - c)(3 - c)$
j	$(t - 1)(t + 2)$	**k**	$(y - 4)(y + 10)$	**l**	$(n - 9)(11 + n)$
m	$(e - 2)^2$	**n**	$(5 + w)^2$	**o**	$(g + 11)^2$

5 Expand $(b + 7)^2$. Select the correct answer **A**, **B**, **C** or **D**.

A $b^2 + 49$ **B** $b^2 + 49b$

C $b^2 + 7b + 49$ **D** $b^2 + 14b + 49$

6 Expand $(x + 8)(3 - x)$. Select **A**, **B**, **C** or **D**.

A $24 - 5x - x^2$ **B** $-x^2 + 24$

C $x^2 + 5x - 24$ **D** $-x^2 + 5x + 24$

EXAMPLE 26

7 Expand each binomial product.

a	$(2x + 5)(x + 3)$	**b**	$(3e + 2)(4e + 5)$	**c**	$(10 + 3p)(p - 1)$
d	$(7d - 2)\,(7d - 2)$	**e**	$(2f - 2)(3f + 5)$	**f**	$(4m - 5)(5 + 3m)$
g	$(3 - 4h)(2 + 5h)$	**h**	$(4p - 5)(2p - 7)$	**i**	$(2m - 3)(4 - 5m)$
j	$(3t + 5)(2t - 1)$	**k**	$(5y - 5)(5y + 5)$	**l**	$(6 - 2a)(2a - 6)$
m	$(3k + 8)^2$	**n**	$(4c - 5)^2$	**o**	$(2d + 9)(3x - 1)$
p	$(5 - 2g)(h + 3)$	**q**	$(7 - 4w)^2$	**r**	$(3n - 5)(4n + 9)$

8 Expand $(3h + 8)(2h - 5)$. Select **A**, **B**, **C** or **D**.

A $6h^2 - 31h - 40$ **B** $6h^2 - h + 40$

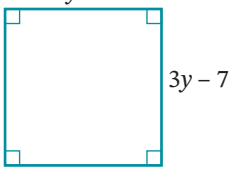
C $6h^2 + 31h - 40$ **D** $6h^2 + h - 40$

9 Write a simplified algebraic expression for the area of each shape.

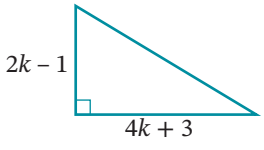
a



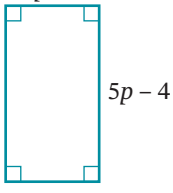
b



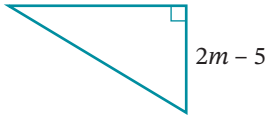
c



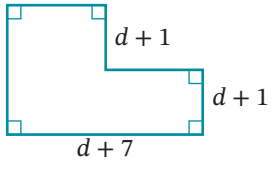
d



e



f



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10 A rectangular barbecue plate has a length of 100 cm and a width of 75 cm. The length and width are both increased by x cm. R C

a Write an expression for the new length of the plate in cm.

b Write an expression for the new width of the plate in cm.

c Hence find a simplified expression for the new area of the plate in cm^2.

d By how much has the area of the plate increased?

e If $x = 0.1$, find the increase in the area of the plate.

11 A family room in a house is to be extended. The room is 4 m long and 3 m wide. The length is to be increased by x m and the width by y m. R C

a Write down expressions for the new length and width in metres.

b Write down a binomial expression for the new area of the room in square metres.

c Expand and simplify your expression for the area.

d Write the algebraic expression for how much the area of the room has increased.

12 Prove that: R C

a $(a - b)^2 = (b - a)^2$

b $(a + b)(a - b) = a^2 - b^2$

c $(a - b)^2 = a^2 - 2ab + b^2$

Power plus ANSWERS ON P. 564

1 If each expression represents an integer, write expressions for the next 3 consecutive integers.

a p **b** $w + 1$ **c** $2n + 1$ **d** $h - 3$

2 Evaluate each expression given $p = 2$, $q = 5$ and $r = -8$.

a $\frac{p+q+r}{3}$ **b** $\frac{pqr}{3}$ **c** $q^2 - 2q$

d $\frac{p-r}{q}$ **e** $\frac{qr}{p}$ **f** $\frac{p}{q} + \frac{r}{q}$

g $(p + q)^2 - 2pq$ **h** $p^3 - q^3 - r^3$ **i** $(p - q)^2 - r^2$

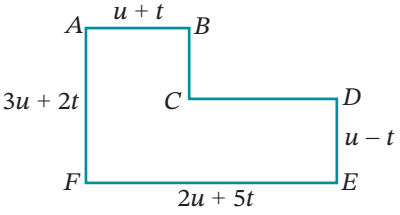
3 For this shape, write an expression for:

a the length of CD

b the length of BC

c its perimeter

d its area



4 Use order of operations to simplify each expression.

a $5 \times 6a \div 3$ **b** $12r \div 2r \times 3$

c $8 \times (6nm \div 3n)$ **d** $10p \times (6pq \div 3q)$

e $15h - 9h \div 3$ **f** $3t + 5 \times 2t$

g $3 \times (x - 5x)$ **h** $(6c + 14c) \div 5$

i $3r + r \times 5 - 1$ **j** $(3r + r) \times (5 - 1)$

5 Expand and simplify each expression.

a $7 - (2m + 5)$ **b** $5x - 3x(2 - x)$

c $5q - 2(3 - 2q) + 5$ **d** $6 - (1 + 3m) - 2m$

e $(2a + 5)^2$ **f** $(y + 1)(y - 1) - (2y - 3)$

6 Find the average of $6r$, $2r + 8$, $r - 5$, $2r$, $r + 7$ and $3r + 8$.

7 A rectangular garden has its longer sides each 5 m longer than its shorter sides. If its longer sides each have a length of y m, then write a simplified expression for:

a the area of the garden **b** the perimeter of the garden.

8 How many hours does it take a car to travel a distance of $20t^2$ kilometres at an average speed of $4t$ km/h?

9 Factorise each expression.

a $8a + 12b - 4$ **b** $20xy + 5y - 10x^2$

c $m^2t + mt + mt^2$ **d** $32 - 24y + 4y^2$

e $12v^2 + 4vw - 16w^2$ **f** $-12 - 9f - 6f^2$

g $\frac{1}{2}p + \frac{1}{2}q$ **h** $\frac{3}{4}k - \frac{1}{4}kt$

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CHAPTER 1 REVIEW

Language of maths

binomial product	common factor	consecutive	difference
distributive law	evaluate	expand	expression
factor	factorise	formula	highest common factor (HCF)
like terms	product	quotient	simplify
substitute	sum	term	variable

1 Why is it possible to add or subtract terms like $6\boldsymbol{n}$ and $4\boldsymbol{n}$ but not $7\boldsymbol{x}$ and $7\boldsymbol{y}$?

2 What word means to rewrite an algebraic expression like $3(\boldsymbol{u} - 5)$ without the grouping symbols?

3 What word means to replace a variable with a number in an algebraic expression?

4 Explain what **product** means.

5 In algebra, what is the opposite of **expand**?

6 Copy and complete: A binomial is an algebraic expression with ____ _____.

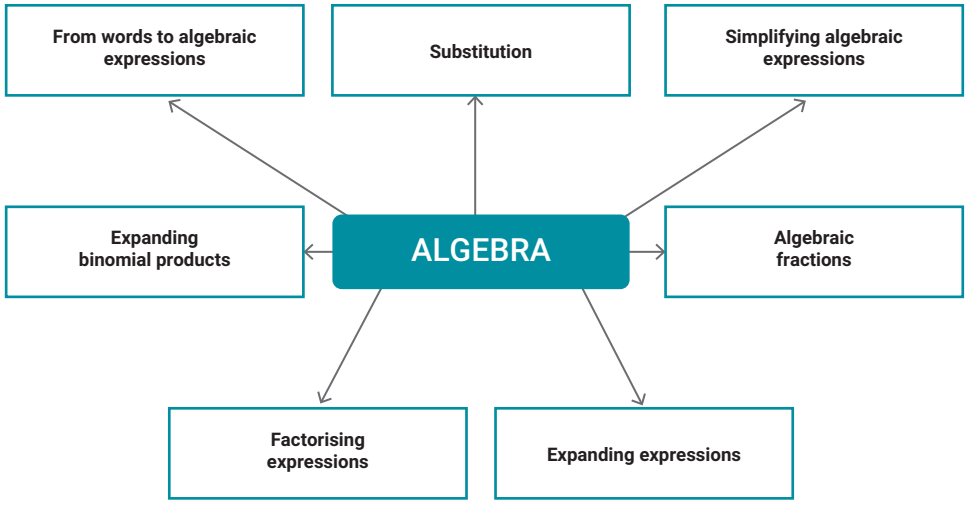
Quiz
Language of maths 1

Puzzle
Algebra crossword

Topic summary

- Write down a list of the new things you have learnt, in your own words.
- What parts of this topic did you like?
- What parts of the topic did you find difficult or not understand?
- Make a list of the skills you have learnt in this topic, such as substituting and expanding.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Worksheet
Mind map: Algebra

Quiz
Test yourself 1

TEST YOURSELF 1

ANSWERS ON P. 564

1.01

1 Convert each statement to an algebraic expression.

a the cost of n pizzas at \$$P$ each

b how many 3s divide into m

c 5 rides at \$$A$ each and 8 rides at \$$B$ each

d decrease \$100 by \$$y$

e the number of minutes in H hours

f the number of litres in m mL

g multiply P by 4 and add 5

h add 5 to T, then multiply by 4

i Q divided by 4, then minus 5

j Q minus 5, then divided by 4



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1.02

2 If $a = 3$ and $b = 2$, evaluate each expression.

a $a + b$ **b** ab **c** $3a + 5b$ **d** $7b - a$

e $8a \div b$ **f** $\frac{a+b}{2}$ **g** $a^2 - b^2$ **h** $2a^2$

1.03

3 Simplify each expression.

a $9y + 7y$ **b** $4ab - 7ab$ **c** $3x^2 + 5x^2$

d $7p + 2q - 5p$ **e** $4w - 5 + 2w$ **f** $8x^2 - 13x - 5x^2$

g $2m + 3n + 5m + 7n$ **h** $5u + 7 - 2u - 3$ **i** $12d^2 + 5d - 7d + 4$

1.04

4 Simplify each expression.

a $9 \times 2n$ **b** $g \times 3h$ **c** $a \times d$

d $(-3) \times 2r$ **e** $12y^2 \div 3y$ **f** $2xy \div y$

g $36a \div 9a$ **h** $15t^2 \div 5t^2$ **i** $20pq \div (-5q)$

1.05

5 Simplify each expression.

a $\frac{h}{6} + \frac{4h}{6}$ **b** $\frac{11k}{8} - \frac{7k}{8}$ **c** $\frac{7}{m} - \frac{5}{m}$

d $\frac{w}{4} - \frac{2w}{5}$ **e** $\frac{4k}{3} - \frac{k}{2}$ **f** $\frac{5}{2d} - \frac{4}{d}$

g $\frac{d}{5k} - \frac{3}{k}$ **h** $\frac{8}{m+1} - \frac{3}{m+1}$

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6 Simplify each expression. 1.06

a $\frac{5}{m} \times \frac{4}{i}$ **b** $\frac{h}{k} \times \frac{3}{k}$ **c** $\frac{14}{3y} \times \frac{9y}{2d}$

d $\frac{r}{4} \div \frac{r}{5}$ **e** $\frac{5}{v} \div \frac{2}{v}$ **f** $\frac{a}{b} \div \frac{a}{b}$

g $\frac{m}{5} \div \frac{5m}{10d}$ **h** $\frac{4}{x} \div \frac{12}{xp}$ **i** $\frac{ad}{6} \times \frac{5a}{2} \div \frac{ad}{8}$

j $\frac{3c}{4d} \div \frac{9c}{16d} \div \frac{6}{8}$ **k** $\frac{15a}{4b} \div \frac{4}{3a^2b} \times \frac{a}{bc}$ **l** $\frac{5y}{8} \div \frac{y}{4} \times \frac{15a}{2y}$

7 Expand and simplify each expression. 1.07

a $5(r - 2)$ **b** $a(a - 10)$ **c** $3y(2x - 5y)$

d $3p(4p - 7q)$ **e** $-6n(n - 9)$ **f** $-6t(t + 7)$

g $b(b + 1) + 5(b + 1)$ **h** $3x(2x - 7) - 5(2x - 7)$ **i** $12 - 6(2x - 3)$

j $5p(2 + 4p - 3q) - (p - q)$ **k** $5(2 - 4p - 3q) - (p + q)$

8 Factorise each expression. 1.08

a $4m - 12$ **b** $xy + xz$ **c** $-3g - 30$

d $16w^2 + 24w$ **e** $-8q + 16$ **f** $15k - 10hk^2$

g $3(y - 6) + y(y - 6)$ **h** $3x(2x - 1) - 2(2x - 1)$ **i** $p(p + 6) - 2(p + 6)$

9 Expand the binomial product $(3d + 5)(d + 4)$ using an area diagram. 1.09

10 Expand each binomial product. 1.09

a $(a + 2)(a + 3)$ **b** $(y + 3)(y - 7)$ **c** $(t - 3)(t + 8)$

d $(h - 5)(h - 4)$ **e** $(4 - g)(3 + g)$ **f** $(3p + 1)(2p + 3)$

g $(5x + 1)(4x - 3)$ **h** $(4r - 5)(3r + 4)$ **i** $(5 - 3q)(2q - 1)$

j $(3m + 5)^2$ **k** $(7 - 2y)^2$ **l** $(3w - 5)(2 - w)$

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